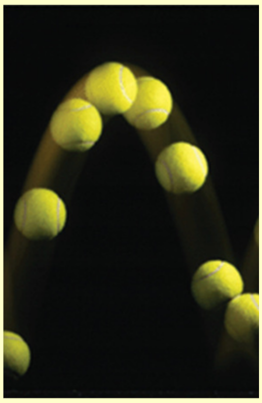
Centre of Mass & Linear Momentum

Centre of mass
Linear momentum
Conservation of momentum
Impulse
Collisions

Centre of Mass



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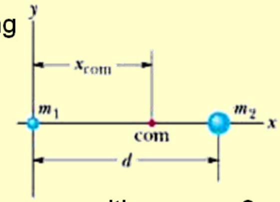
Centre of Mass (com)

- com of a system of particles is the point that moves as though
  (1) all the system's mass were concentrated there and
  (2) all external forces were applied there.
- Where is the com?
  - Start with a small system of only 2 particles, separated by d.
  - Choose a set of axes with (0,0) passing through m_1.
  - Define com of the 2 particle system:

$$x_{com} = \frac{m_2}{m_1 + m_2} d$$



  - If $m_2 = 0$, then m_1 and com must lie in same position: $x_{com} = 0$
  - If $m_1 = 0$, then m_2 and com must lie in same position: $x_{com} = d$.
  - If $m_1 = m_2$, then com must be halfway between: $x_{com} = \frac{1}{2}d$.
  - If $m_1 \neq 0$, $m_2 \neq 0$, then com $0 < x_{com} < d$.

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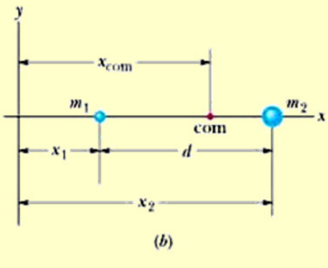
Centre of Mass (com)



- Where is the com?
  - Same 2 particles system, separated by d.
  - Choose a set of axes (general).
  - Define com of the 2 particle system:

$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

  - If $x_1 = 0$, then $x_2 = d$ equation reduces to same as previous
  - com is independent of coordinate system.
  - $x_{com} = \frac{m_1 x_1 + m_2 x_2}{M}$ M is the total mass of system
  - For general system of n particles along x axis:

$$x_{com} = \frac{m_1 x_1 + m_2 x_2 + \ldots + m_n x_n}{M} = \frac{1}{M} \sum_{i=1}^{n} m_i x_i$$

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Centre of Mass (com)

- Where is the com in 3-D?
  - For general system of n particles in 3 dimensions:

$$x_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i$$

$$y_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i y_i$$

$$z_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i z_i$$

  - $\vec{r}_{com} = x_{com}\hat{i} + y_{com}\hat{j} + z_{com}\hat{k}$

$$\vec{r}_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i \vec{r}_i$$

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Question

Three objects are located in the x-y plane as shown in the figure. Determine the x coordinate of the center of mass for this system of three objects. Note the masses of the objects: m_A = 6.0 kg, m_B = 2.0 kg, and m_C = 4.0 kg.

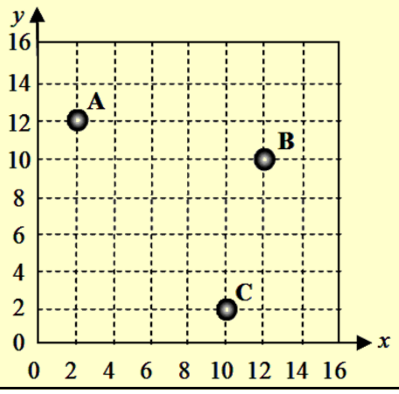
a) 5.6

b) 6.3

c) 7.6

d) 8.3

e) 8.9



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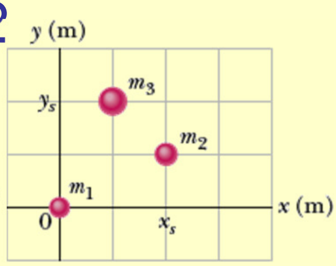
Problem 2



- The figure shows a 3 particle system with masses m_1 = 3.0 kg, m_2 = 4.0 kg and m_3 = 8.0 kg. The scales on the axes are set by x_s = 2.0 m and y_s = 2.0 m.
(a) What are the x coordinate and the y coordinate of the system's centre of mass?
(b) If m_3 gradually increases, how will the com shift in relation to m_3.

DO SAMPLE PROBLEM 9-1

Linear Momentum

- Linear Momentum: A vector quantity $\vec{p}$ defined as:

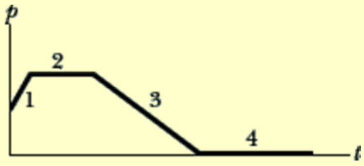
$$\vec{p} = m\vec{v}$$

- m is the mass and $\vec{v}$ is the velocity of the particle
- Momentum has the same direction as the velocity
- Units: kg·m/s
- Time rate of change of momentum of a particle = net force acting on particle and is in the direction of the force

$$\vec{F}_{net} = \frac{d\vec{p}}{dt} = \frac{dm\vec{v}}{dt} = m\frac{d\vec{v}}{dt}, \quad \text{Newton's second law}$$

- The net external force changes the particles linear momentum, or
- Momentum can be changed only by a net ext force

Checkpoint 3



- The figure gives the magnitude p of the linear momentum versus time for the particle moving along an axis. A force directed along the axis acts on the particle.
  - Rank the four regions indicated according to the magnitude of the force, greatest to smallest.
  - In which region is the particle slowing?

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Linear momentum of a System of Particles

- Extend the definition of linear momentum to a system of n particles, each having its own mass, velocity and linear momentum.
- Total linear momentum $\vec{P}$

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \ldots + \vec{p}_n$$
$$= m\vec{v}_1 + m\vec{v}_2 + m\vec{v}_3 + \ldots + m\vec{v}_n$$
$$= M\vec{v}_{com}$$
$$\vec{F}_{net} = M\frac{d\vec{v}_{com}}{dt} = M\vec{a}_{com}$$

- If $\vec{F}_{net} \neq 0$, $\vec{P}$ will change

  If $\vec{P}$ changes, $\vec{F}_{net}$ must be exerted

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Question

- The momentum of an object is not dependent on which one of the following quantities?

  a) acceleration

  b) mass

  c) speed

  d) velocity

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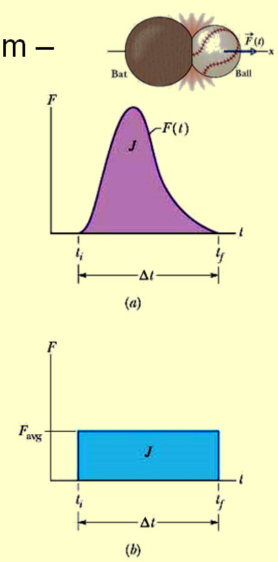
Collision and Impulse

- In order to change linear momentum – a force must act on a object.
- Body colliding with a baseball bat:
  - External force on object is brief
  - Large magnitude
  - Momentum changes.
- $\vec{F} = \dfrac{d\vec{P}}{dt}$

$$\vec{F}dt = d\vec{p}$$

$$\int_{p_i}^{p_f} d\vec{p} = \int_i^f \vec{F}dt$$

$$\Delta p = \int_i^f \vec{F}dt = \vec{J}$$ IMPULSE



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Question

- Complete the following statement: The change in momentum with respect to time for an object

a) is always positive.

b) is always equal to zero.

c) has a constant value if the force acting on the object is conservative.

d) is equal to the net force acting on the object.

e) is a scalar quantity that may be negative, zero, or positive.

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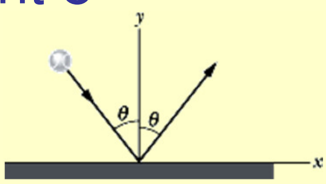
Checkpoint 5



- The figure shows an overhead view of a ball bouncing from a vertical wall without any change in its speed. Consider the change Δp in the ball's linear momentum.
  (a) Is Δp_x positive, negative or zero?
  (b) Is Δp_y positive, negative or zero?
  (c) What is the direction of Δp?

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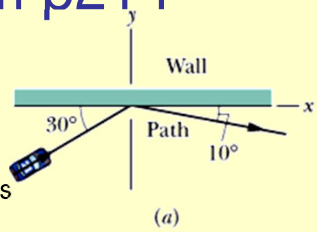
Sample Problem p214



- The figure shows an overhead view of the path taken by a race car driver as his car collides with the track wall. Just before the collision he is travelling at speed v_i = 70 m/s along a straight line at 30° from the wall.
  Just after the collision he is travelling at speed v_f = 50 m/s along a straight line at 10° from the wall. His mass m = 80 kg.
  (a) What is the impulse on the driver due to the collision?
  (b) If the collision lasts for 14 ms, what is the magnitude of the average force on the driver during the collision?

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Sample Problem p214

- (b) If the collision lasts for 14 ms, what is the magnitude of the average force on the driver during the collision?

What would the effect be if the collision was made to last longer?

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Conservation of Linear Momentum

- In a closed (nothing enters or leaves), isolated system
- $\vec{F}_{net} = 0$ (isolated)

  $\frac{d\vec{P}}{dt} = 0$

  $\vec{P} =$ constant
- $\vec{P}_i = \vec{P}_f$ - Law of conservation of linear momentum

- If the component of the net external force on a closed system is zero along an axis, then the component of the linear momentum of the system along that axis cannot change.

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Checkpoint 6

- An initially stationary device lying on a frictionless floor explodes into two pieces, which then slide across the floor. One piece slides in the positive direction of an x axis.
  (a) What is the sum of the momenta of the two pieces after the explosion?
  (b) Can the second piece move at an angle to the x axis?
  (c) What is the direction of the momentum of the second piece?

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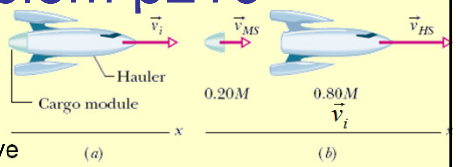
Sample Problem p216



- The figure shows a space ship and a cargo module, of total mass M, travelling along an x axis in space. They have an initial velocity $\vec{v}_{rel}$ of 2100 km/h relative to the Sun. With a small explosion, the cargo module is ejected (mass of 0.20M). The space ship then travels 500 km/h faster than the module along the x axis. The relative speed $\vec{v}_{ss}$ between the hauler and the module is 500 km/h. What is the velocity of the space ship relative to the Sun?

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Momentum and K E

- Previous sections – considered the momentum of two particle-like bodies colliding, but focused on only one – e.g. ball in bat and ball collision.
- Now will focus on the system – i.e. bat and the ball
- If $\vec{F}_{net} = 0$, then $\Delta\vec{P} = 0$
  Need not know details of collision.
- When 2 objects collide, and we consider the KE before and after the collision:
  - If $\Delta K = 0$, the collision is an elastic collision
  - If $\Delta K \neq 0$, the collision is an inelastic collision
  - Inelastic collisions involve a loss in the KE of the system.

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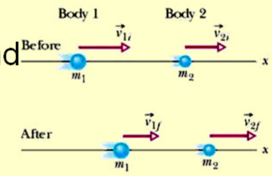
Inelastic collisions – 1-D

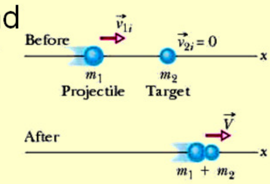


- Figure shows 2 bodies just before and just after they have a 1-D inelastic collision
- System = 2 bodies
- Closed and isolated
- $\vec{P}_i = \vec{P}_f$

  $\vec{p}_{1i} + \vec{p}_{2i} = \vec{p}_{1f} + \vec{p}_{2f}$

- In 1 dimension: $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$

Inelastic collisions – 1-D

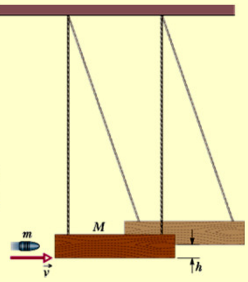


- Figure shows 2 bodies just before and just after they have a 1-D completely inelastic collision
- If one body is moving and the other is stationary – projectile and target
- In 1 dimension:

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

- Since $\frac{m_1}{m_1 + m_2} < 1$, v_f will be less than v_{1i}
- COM of system continues at constant velocity

Sample Problem 9-9



- The ballistic pendulum was used before electronic timing devices. The figure shows a large block of wood of mass M = 5.4 kg hanging from two long cords. A bullet of mass m = 9.5 g is fired into the block, coming to rest in the block. The block-bullet then swing upward, their centre of mass rising a vertical distance h = 6.3 cm before the pendulum comes to rest for a moment. What is the speed of the bullet just prior to the collision?

How does conservation of mechanical energy relate to 2 different objects where other types of energy is also involved?

Split this complicated motion into 2 steps which are analysed separately:
(1) The bullet-block collision
(2) Bullet-block rise during which mechanical energy is conserved.

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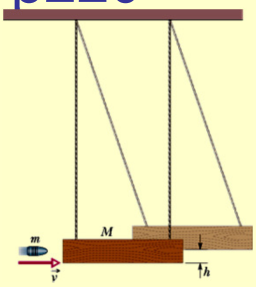
Sample Problem p220



- M = 5.4 kg
  m = 9.5 g
  h = 6.3
- What is the speed of the bullet just prior to the collision?

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Elastic Collisions – 1D

- Can approximate some everyday collisions to be elastic – e.g. pool balls colliding
- $\begin{pmatrix}\text{total } KE \text{ before} \\ \text{collision}\end{pmatrix} = \begin{pmatrix}\text{total } KE \text{ after} \\ \text{collision}\end{pmatrix}$
- $\begin{pmatrix}\text{linear momentum} \\ \text{before collision}\end{pmatrix} = \begin{pmatrix}\text{linear momentum} \\ \text{after collision}\end{pmatrix}$
- In an elastic collision, the KE of the individual objects can change, but the total KE of the system does not change.

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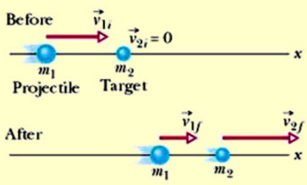
Elastic Collisions – stationary target



- Figure shows 2 bodies just before and just after they have a 1-D collision
- Projectile of m_1, initial v_{1i}, moves towards
- Target of m_2, initial at rest, $v_{2i} = 0$
- Closed, isolated system
- $$m_1 v_{1i} = m_1 v_{1f} + m_2 v_{2f}$$
  $$½ m_1 v_{1i}^2 = ½ m_1 v_{1f}^2 + ½ m_2 v_{2f}^2$$
- Special conditions – read on own p 222

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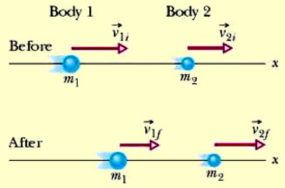
Elastic Collisions – moving target

- Figure shows 2 bodies just before and just after they have a 1-D collision
- Projectile of m_1, initial v_{1i}, moves towards
- Target of m_2, initial v_{2i}
- Closed, isolated system
- $$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$
  $$½m_1 v_{1i}^2 + ½m_2 v_{2i}^2 = ½m_1 v_{1f}^2 + ½m_2 v_{2f}^2$$



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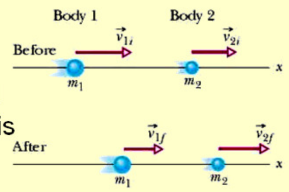
Checkpoint 8

- What is the final linear momentum of the target in the figure if the initial linear momentum of the projectile is 6 kg·m/s and the final linear momentum of the projectile is
  (a) 2 kg·m/s and
  (b) -2 kg·m/s?
  (c) What is the final KE of the target if the initial and final KE of the projectile are 5 J and 2 J, respectively?



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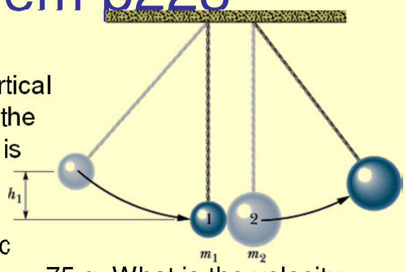
Sample Problem p223



- Two metal spheres, suspended by vertical cords, initially just touch, as shown in the figure. Sphere 1, with mass m_1 = 30 g is pulled to the left to a height h_1 = 8.0 cm and then released from rest. After swinging down, it undergoes an elastic collision with sphere 2, whose mass m_2 = 75 g. What is the velocity v_{1f} of sphere 1 just after the collision?
- Split problem into 2 steps, similar to ballistic pendulum problem
- Let lowest level be PE = 0.

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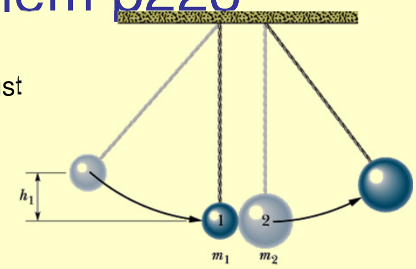
Sample Problem p223



- What is the velocity v_{1f} of sphere 1 just after the collision?

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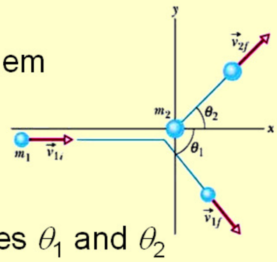
Collision in 2-D



- 2 bodies collide – impulse between them determine the direction of travel.
- $\vec{P}_{1i} + \vec{P}_{2i} = \vec{P}_{1f} + \vec{P}_{2f}$

  $K_{1i} + K_{2i} = K_{1f} + K_{2f}$
- Two bodies move off at different angles θ_1 and θ_2
- x components of momentum:

  $m_1 v_{1i} = m_1 v_{1f} \cos\theta_1 + m_2 v_{2f} \cos\theta_2$
- y components of momentum:

  $0 = m_1 v_{1f} \sin\theta_1 + m_2 v_{2f} \sin\theta_2$
- Kinetic Energy of system

  $½ m_1 v_{1i}^2 = ½ m_1 v_{1f}^2 + ½ m_2 v_{2f}^2$

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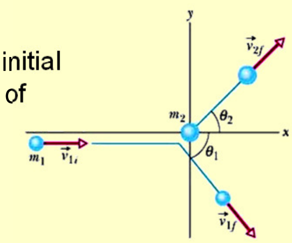
Checkpoint 9



- In the Figure, suppose the projectile has an initial momentum of 6 kg·m/s, a final x component of momentum of 4 kg·m/s, and a final y component of momentum of -3 kg·m/s .
  For the target, what then are
  (a) the final x component of momentum and
  (b) the final y component of momentum?

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Problem 75

- A projectile proton with a speed of 500 m/s collides elastically with a target proton initially at rest. The two protons then move along perpendicular paths, with the projectile path at 60° from the original direction. After the collision, what are the speeds of (a) the target proton and (b) the projectile proton?

Problem 75

- After the collision, what are the speeds of (a) the target proton and (b) the projectile proton?